

Ethics of AI Agents in Biomedical Data Science: A Framework for Responsible Use of LLM-Based Agentic Systems

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Abstract

Large language model (LLM) based AI agents are increasingly deployed in biomedical data science to assist with literature review, code generation, data analysis, and manuscript preparation. Unlike single-turn chatbot interactions, agentic workflows involve planning, tool use, persistent memory, and multi-agent coordination, substantially expanding the ethical surface area of AI-assisted research. This article presents a comprehensive framework for the responsible use of AI agents in biomedical contexts. The framework rests on a positional metaethics: ethical commitments are defended through public justification rather than derived from empirical measurement. Four such commitments organize the argument: non-delegable human accountability, structural verification, sociodemographic categories treated as historically constructed rather than as stable natural kinds, and multi-agent critique as conditional rather than magical. From this grounding the framework develops two foundational concepts. First, we introduce *invalidation* as a broader and more actionable concept than “hallucination,” encompassing factual, logical, normative, and structural breaches of constraints. Second, we examine multi-agent critique architectures (such as “flaws-of-others” loops) that can reduce invalidation rates while introducing new ethical tradeoffs involving privacy, compute burden, and epistemic homogenization. We map seven categories of ethical risk: epistemic propagation, accountability and authorship, provenance and reproducibility, confidentiality, bias and fairness, security, and sustainability. For each category, we provide concrete mitigation strategies aligned with Responsible Conduct of Research (RCR) principles and current publication ethics guidance from COPE, ICMJE, and WAME. We conclude with a governance-by-design checklist and disclosure templates suitable for immediate adoption in biomedical research workflows.

Keywords: AI agents; research integrity; invalidation; hallucination; provenance; confidentiality; authorship; bias; sustainability; biomedical data science; responsible conduct of research

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1 Introduction

1.1 Two kinds of claim about the world

A scientist can weigh a cup. Place it on a calibrated scale, record the reading, repeat. Any careful observer reaches the same value within a known tolerance, and the uncertainty itself is measurable. A scientist cannot weigh a deity. One may affirm or deny such a claim, but it does not become reproducibly adjudicable by adding more decimal places. The boundary that separates these two cases runs not between science and religion or between technical and non-technical subjects, but between claims that admit public, reproducible procedures of correction and claims that depend on a position one occupies. In a biomedical workflow this takes concrete form: a cohort count, a code execution, and a cited reference can each be checked against a protocol or a source; whether an observed privacy risk is acceptable for a secondary-use study cannot, and requires defensible commitments about consent, equity, and accountability.

For the purposes of biomedical data-science practice, empirical knowledge will mean claims made accountable to reproducible procedures of measurement, computation, or audit. Measurement so understood is not value-free, but it is intersubjectively checkable; a normative claim is not, though it is adopted, defended, and made binding through institutions. Hume drew this line in *A Treatise of Human Nature* (1739–1740): one cannot derive an “ought” from an “is” without smuggling in a premise that no observation supplies [1]. Moore’s “naturalistic fallacy” in *Principia Ethica* (1903) gave the gap a name and an argument: any attempt to define a moral predicate in purely natural terms leaves an open question that the definition cannot close [2]. Hume and Moore establish the non-reduction of normative judgment to empirical description. They do not by themselves settle whether normative claims can have universal validity. The positional stance defended below is an additional metaethical commitment, introduced in Section 1.3.

1.2 Kant’s two questions, in biomedical terms

Kant adds a related but distinct move. The *Critique of Pure Reason* [3] asks under what conditions theoretical knowledge is possible. The *Critique of Practical Reason* [4] asks how practical reason can guide action through what Kant calls the “moral law.” Kant himself defends the universality of that moral law and is not a positivist. The present manuscript draws on Kant only to distinguish modes of justification: empirical claims require reproducible evidential procedures, whereas normative claims require defensible practical commitments. Both modes are unavoidable in biomedical research.

This distinction structures the taxonomy of invalidation that Section 3 develops. Factual and structural invalidations belong primarily to the domain of what can be checked by procedure: a fabricated citation can be tested against a publisher’s resolver, and a malformed schema can be validated against a contract. Normative invalidation belongs primarily to the domain of what ought to be done: an output may be factually accurate yet impermissible because it exposes protected health information, reinforces stigmatizing categories, or violates a data-use agreement. Logical invalidation polices the bridge between the two, identifying invalid inferences by which descriptive findings are made to support practical conclusions they do not warrant.

1.3 Positionality without naive relativism

If normative claims do not admit reproducible adjudication, where do they come from? The metaethical stance adopted here is *practical positionality*: ethical commitments are articulated from within institutions, histories, communities, and material conditions, and they remain open to public justification and ongoing critique. Haraway [5] names the alternative the “god-trick”: the pretense of a view from nowhere that delivers values without a knower. Agentic AI systems often present outputs as if generated from nowhere; they have not. Every output bears the marks of training data, vendor incentives, and the position of the institution deploying the system.

Positionality is not naive relativism. Williams [6] distinguishes two questions an ethics may be asked to answer: whether values can be argued for, and whether they have universal warrant. The stance adopted here answers yes to the first and abstains on the second. The authority of biomedical research values in practice comes through public justification, institutional adoption, legal force, and ongoing critique. The research community already operates this way: informed consent and conflict-of-interest disclosure are positional commitments adopted, debated, and made binding through professional codes and statute.

Contemporary AI-ethics literature offers competing attempts to operationalize values under technical and institutional constraints. Floridi and colleagues [7] propose the *AI4People* framework of five principles; Mittelstadt and colleagues [8] earlier mapped the ethics of algorithms across distinct types of concern; Whittlestone and colleagues [9] argue that principle lists, read as context-free axioms, conceal the tensions among them. Bender and colleagues [10] foreground training-data biases, scale-induced harms, and the labor and environmental costs of large language models; Crawford [11] situates AI within political economy and planetary extraction; Birhane [12] argues that algorithmic injustice requires relational, contextual ethics; and Vallor [13] defends a virtue-ethics approach that locates values in cultivated dispositions rather than in axioms. These accounts converge on the practical point that technical systems carry social commitments even when their interfaces make those commitments invisible.

1.4 Regulation as the operational form of positional values

Positional values become binding through governance, which operates at different levels of authority. The U.S. NIST AI Risk Management Framework [14] is voluntary risk-management guidance. The European Union Artificial Intelligence Act [15] is binding risk-based regulation with extraterritorial reach. The UNESCO Recommendation on the Ethics of Artificial Intelligence [16] is a global standard-setting instrument grounded in human-rights commitments and applied to member states through national implementation. The OECD Council Recommendation on Artificial Intelligence [17] comprises intergovernmental principles adopted in 2019 and updated in 2024. Their differences are not defects to be eliminated by a better benchmark. Each is a serious attempt to specify which harms matter, which actors bear responsibility, and which forms of evidence count for trustworthy AI. The choice among them, and the choice of how to operationalize them in a specific research workflow, is itself a value choice.

1.5 The positional commitments of this manuscript

The manuscript adopts four positional commitments as starting points, stated here so the reader meets the metaethics before the technical content, and defended in what follows.

First, *human accountability for AI-assisted research is non-delegable*. Agents are tools, not co-bearers of responsibility. The International Committee of Medical Journal Editors [18], the Committee on Publication Ethics [19], and the World Association of Medical Editors [20] converge on the position that AI tools cannot meet the authorship criterion of accountability; the National Institutes of Health applies the same logic to peer review [21].

Second, *verification is structural rather than optional*. Factual claims, generated code, retrieved citations, and statistical interpretations require planned checks rather than post-hoc reassurance. The discursive-network framework [22] provides the scaffolding for what verification means when agents produce statements that propagate through manuscripts, code repositories, and the published record.

Third, *sociodemographic categories used in biomedical data are unstable across time and place and require explicit definition, provenance, and sensitivity analysis rather than assumed comparability*. The 2024 revision of U.S. Office of Management and Budget Statistical Policy Directive No. 15 [23] is the most recent reminder that even federal categories of race and ethnicity are historically constructed rather than stable natural kinds.

Fourth, *multi-agent critique is conditional rather than magical*. Under stated conditions it can reduce some classes of invalidation [22], but it can also amplify privacy exposure across services, propagate correlated errors across model families, and produce a false sense of consensus.

These four are positional values: defensible, but not universal warrants.

1.6 Contributions and roadmap

The shift from single-turn chatbot interaction to agentic systems (workflows that plan multi-step tasks, invoke external tools, maintain memory across interactions, and coordinate with other agents [24–27]) multiplies the ethical surface of AI-assisted research. When an agentic system autonomously retrieves literature, generates analysis code, executes that code on sensitive data, and drafts results text, the four positional commitments above engage at every step. The agent becomes a participant in a *discursive network* where statements can propagate, mutate, and acquire authority even when incorrect [22]. The remainder of the manuscript develops the framework needed to operationalize the commitments under those conditions.

Our contributions are as follows:

1. We extend the invalidation framework of Gutiérrez [22] from three categories (factual, logical, structural) to four by adding normative invalidation, and explain why this taxonomy matters for research integrity.
2. We analyze the ethical tradeoffs of multi-agent critique architectures, demonstrating how verification loops can reduce invalidation while introducing new risks of privacy amplification, compute burden, and epistemic homogenization.

3. We present a systematic risk map covering seven categories of ethical concern specific to agentic workflows in biomedical contexts.
4. We provide governance-by-design principles, including a practical checklist and disclosure templates aligned with current publication ethics standards [18–20].

The article is structured as follows. Section 2 formalizes the agentic-AI landscape in which the four commitments must operate, drawing on the recent surveys of [28], [26], and [27], and on the discursive-network framework of [22]. Section 3 develops the invalidation taxonomy. Section 4 analyzes multi-agent critique architectures. Section 5 presents the seven-category risk map. Section 6 offers governance principles and practical tools. Section 7 traces an end-to-end illustrative workflow. Section 8 discusses implications and limitations. Section 9 concludes.

2 Background: AI Agents and Discursive Networks

2.1 Defining AI Agents

For the purposes of this article, we define an **AI agent** as an LLM-centered system that exhibits four capabilities:

- (a) **Planning:** The ability to interpret high-level goals and decompose them into sequences of subtasks, potentially with branching logic and error recovery.
- (b) **Tool use:** The ability to invoke external capabilities such as web search, code execution, database queries, file manipulation, and API calls.
- (c) **Memory:** The ability to maintain state across interaction steps, whether through explicit memory stores, scratchpads, or retrieval-augmented mechanisms.
- (d) **Multi-agent interaction:** The ability to communicate with humans and other agents, delegate subtasks, aggregate results, and participate in collaborative or adversarial workflows.

This definition encompasses a spectrum of systems, from relatively simple tool-augmented chatbots to complex multi-agent orchestrations. The key distinction from single-turn LLM use is that agents can take *autonomous actions with persistent effects*, creating audit trails that extend beyond a single prompt-response pair.

Recent surveys have documented the rapid proliferation of agent architectures [26, 27, 29]. Agents have been applied to software engineering, scientific research, data analysis, and administrative tasks. In biomedical contexts, agents are being deployed for literature triage, clinical note summarization, statistical code generation, and even elements of peer review assistance [24].

An AI agent in the sense used in this article is an LLM-centered system that exhibits at minimum planning, tool use, and memory. Multi-agent interaction extends the architecture and adds a distinct ethical surface, addressed separately in Section 4.

2.2 The Discursive Network Framework

The discursive network framework, drawn from recent work on the circulation of statements in human-AI communication systems, treats a research workflow as a graph in which statements move among nodes (humans, AI systems, documents). The framework makes explicit a property usually left implicit in discussions of AI hallucination: an inaccurate statement, once entered into the workflow, has effects that depend not only on its accuracy but on how it propagates. Three of those propagation properties are operationally important for agent ethics and are summarized below.

Our ethical analysis draws on the *discursive network* framework developed by Gutiérrez [22]. In this framework, communication systems are modeled as networks in which statements (propositions, claims, data) circulate among nodes (humans, AI systems, documents, databases). Statements can be generated, transmitted, transformed, and stabilized as they move through the network.

Three properties of discursive networks are particularly relevant to AI agent ethics:

1. **Circulation:** Statements produced by agents can propagate through research workflows, entering manuscripts, code repositories, presentations, and eventually the published literature. Once in circulation, statements acquire lives of their own.
2. **Authority accumulation:** Statements gain credibility through association with trusted sources, repetition across contexts, and integration into larger narratives. An agent-generated claim that appears in a peer-reviewed publication acquires authority independent of its origin.
3. **Mutation:** As statements move through networks, they can be paraphrased, summarized, combined with other statements, and stripped of context. Agent outputs often undergo such transformations before reaching their final form.

This framework illuminates why agent ethics cannot be reduced to the question of whether individual outputs are “correct.” Even if most agent outputs are accurate, the circulation dynamics of discursive networks mean that errors can propagate widely and prove difficult to retract. Moreover, the fluid authorship of agent-assisted content complicates traditional mechanisms of scientific accountability.

2.3 The Expanded Ethical Surface

Single-turn LLM interactions present ethical challenges that have been extensively documented: generation of false information (“hallucination”), reproduction of biased patterns from training data, potential for misuse, and environmental costs of inference [10, 30, 31]. Agentic workflows inherit all these concerns while adding new dimensions:

- **Scope of action:** Agents can take actions with real-world effects, such as executing code that modifies files, querying databases, or sending communications. Errors compound across action sequences.
- **Opacity of reasoning:** Multi-step agent workflows produce intermediate states that may not be visible to users, making it difficult to audit how conclusions were reached.

- **Persistence:** Agent memory and state can carry forward errors or biases across sessions, potentially affecting multiple projects or users.
- **Coordination complexity:** Multi-agent systems can exhibit emergent behaviors not present in individual components [28], including collective biases and cascading failures.

These expanded risks motivate the need for a correspondingly expanded ethical framework.

3 Invalidation: Beyond Hallucination

3.1 The Limitations of “Hallucination”

The term “hallucination” has become standard in discussions of LLM reliability, typically referring to the generation of false or fabricated information presented as fact [32–34]. While useful, this framing has significant limitations for research ethics:

1. **Narrow scope:** Hallucination typically denotes factual errors (false claims about the world), but agents can fail in other ways: producing logically inconsistent outputs, violating ethical norms, or generating structurally malformed results.
2. **Connotation of randomness:** The term suggests sporadic, unpredictable failures, obscuring the systematic patterns in how LLMs err.
3. **Focus on generation:** Hallucination centers on what the model produces, rather than on the broader workflow in which outputs circulate and acquire consequences.

3.2 A Taxonomy of Invalidation

Building on the invalidation framework of Gutiérrez [22], which identifies factual, logical, and structural breaches, we extend the taxonomy to include a fourth category, **normative invalidation**, which captures breaches of ethical, legal, or policy requirements. The full four-type taxonomy is more complete for biomedical contexts, where regulatory and ethical constraints carry as much weight as factual accuracy. This framing emphasizes that validity is always relative to a set of requirements, and that different requirements apply in different contexts.

We distinguish four types of invalidation:

Factual invalidation. The output contradicts established facts about the world. Examples include fabricated citations (references to papers that do not exist), incorrect statistical claims, and false assertions about biomedical mechanisms. Factual invalidation corresponds most closely to traditional notions of hallucination.

Logical invalidation. The output contains internal contradictions or invalid inferences. Examples include statistical non sequiturs (conclusions that do not follow from the analysis), inconsistent claims within a single document, and circular reasoning. Logical invalidation may co-occur with factually correct individual statements.

Normative invalidation. The output violates ethical, legal, or policy requirements. Examples include exposure of protected health information, biased characterizations of populations, and violations of informed consent provisions. Normative invalidation concerns what outputs *should not* contain, regardless of factual accuracy.

Structural invalidation. The output fails to satisfy format or schema requirements. Examples include malformed JSON, code that does not compile, broken hyperlinks, and outputs that exceed length constraints. Structural invalidation is particularly relevant in agentic workflows where outputs serve as inputs to downstream tools.

This taxonomy is not exhaustive, and the categories can overlap. A fabricated citation (factual invalidation) might also violate publication ethics (normative invalidation). Nevertheless, the taxonomy provides a more complete vocabulary for identifying and addressing agent failures.

When a single failure can be classified as more than one type of invalidation, classify it by the type of constraint that fails first in the verification workflow appropriate to the output. A fabricated citation fails the factual constraint (the cited paper does not exist) before any normative constraint engages, and is therefore classified as factual. A correctly cited but contextually inappropriate paper that mischaracterizes a vulnerable population fails the normative constraint and is classified as normative. This rule does not deny the multi-type nature of the failure; it identifies the most actionable verification target.

3.3 Implications for Research Integrity

The invalidation framework has direct implications for research integrity:

1. **Verification must be multidimensional.** Checking factual accuracy is necessary but not sufficient; logical coherence, normative compliance, and structural correctness also require attention.
2. **Different outputs require different verification strategies.** A literature review requires fact-checking citations; a statistical analysis requires logical verification of the inference chain; a data processing pipeline requires structural validation of outputs at each stage.
3. **Invalidation rates vary by task and domain.** Biomedical content may have different invalidation profiles than other domains due to specialized terminology, complex causal relationships, and stringent regulatory requirements.

The mathematical framework in Gutiérrez [22] argues that invalidation should be treated as a structural risk in complex LLM-mediated workflows: under stated model assumptions, a residual error rate is unavoidable. Whether a specific task, architecture, or domain reaches a given empirical invalidation rate remains an empirical question. The ethical imperative is therefore not to eliminate invalidation (which the framework argues is generally infeasible) but to detect, bound, and mitigate its propagation through research workflows.

4 Multi-Agent Critique: Benefits and Tradeoffs

4.1 The Verification Advantage

A key insight from research on LLM reliability is that *verification is often easier than generation* in this class of models [35, 36]. An LLM that generates a false claim may nevertheless recognize that claim as false when asked to evaluate it critically. This asymmetry motivates architectures in which multiple agents review each other’s outputs.

The “flaws-of-others” (FOO) pattern provides a structured approach [22]. In this architecture, multiple agents independently produce outputs for a task; each agent then critiques the outputs of the others; a harmonizing agent synthesizes the critiques and produces a refined result. Empirical studies of multi-agent debate and self-critique architectures [37, 38] report reductions in factual and reasoning errors relative to single-model baselines, with magnitudes that vary by task, model family, and prompt design. The effect should therefore be treated as a useful but not guaranteed verification gain.

For instructional and operational use, the discursive-network framework of [22] yields a closed-form expression for the steady-state share of false statements with n mutually detecting agents:

$$\pi_F^{(n)} = \frac{p}{p + q + (n - 1)d}, \quad (1)$$

where p is the per-agent invalidation rate, q is the self-repair rate, and d is the cross-detection probability. Setting $\pi_F^{(n)} \leq \epsilon$ and solving for n gives the minimum number of agents required to reach a target error tolerance:

$$n_{\min} = \left\lceil 1 + \frac{p\left(\frac{1}{\epsilon} - 1\right) - q}{d} \right\rceil. \quad (2)$$

The reference values $q = 0.05$ and $d = 0.19$ from [22] are appropriate starting points for biomedical workflows; both should be re-estimated for a given task and ensemble before being relied upon for quantitative agent-count decisions.

4.2 Ethical Tradeoffs of Multi-Agent Architectures

While multi-agent critique can improve reliability, it introduces new ethical considerations:

Privacy amplification. Each additional agent that processes a piece of text is another potential vector for privacy leakage. If the original prompt contains sensitive information (patient data, unpublished research, confidential peer review content), sharing that content with multiple agents multiplies exposure risk. Research on membership inference attacks has demonstrated that language models can leak information about their inputs [39, 40].

Compute and sustainability costs. Multi-agent workflows require multiple inference passes, often scaling quadratically in the number of agents if each critiques every other. This increases both financial costs and

environmental impact [41, 42]. When “best practices” require extensive multi-agent verification, resource constraints may create inequities between well-funded and resource-limited research groups.

Epistemic homogenization. When agents trained on similar data review each other, they may converge on outputs that reflect shared biases rather than ground truth. The critique process may suppress minority-but-correct perspectives in favor of bland consensus. This risk is particularly acute when all agents in a workflow come from the same model family or vendor.

Attribution diffusion. When multiple agents contribute to an output, determining responsibility for errors becomes more complex. The harmonized result may contain elements from several agents, making it difficult to trace the origin of specific claims.

4.3 Design Principles for Responsible Multi-Agent Use

Given these tradeoffs, we propose several design principles:

1. **Right-size verification:** The number of critique agents should be proportional to the stakes of the task. Routine formatting checks need not involve the same verification depth as claims that will appear in peer-reviewed publications.
2. **Architectural diversity:** Where feasible, multi-agent workflows should include agents from different model families to reduce correlated biases.
3. **Hierarchical verification:** Inexpensive checks (schema validation, unit tests, retrieval verification) should precede expensive multi-agent critique, reducing overall compute burden.
4. **Explicit role assignment:** At least one agent should be tasked with identifying potential false negatives: claims that the consensus may have wrongly accepted.

5 Ethical Risk Map for Agentic Workflows

This section presents a systematic map of ethical risks in agentic biomedical workflows. For each risk category, we describe the concern, provide examples, and offer concrete mitigation strategies.

5.1 Epistemic Risk: Invalidation Propagation and Overtrust

Concern. Agent-generated text can be difficult to distinguish from expert-authored prose. When invalid statements enter circulation, they can propagate through documents, code, presentations, and eventually the published literature. The fluent style of LLM outputs may engender overtrust, leading researchers to accept claims without adequate verification [43].

Examples. A literature review agent fabricates citations that a researcher incorporates without checking. An analysis agent makes a statistical claim (“the effect is significant at $p < 0.01$ ”) that the researcher propagates to a manuscript without verifying the underlying calculation.

Mitigations.

- Require traceable sources (DOIs, URLs) for all factual biomedical claims; reject outputs with “phantom citations.”
- Implement retrieval-augmented verification: before accepting factual claims, retrieve supporting evidence from trusted sources.
- Establish a verification budget: define which output categories require independent verification, by whom, and using what methods.
- Maintain skepticism proportional to stakes: claims destined for publication require more stringent verification than internal working notes.
- Sampling-based self-consistency methods such as SelfCheckGPT [44] provide a cheap pre-screen for factual inconsistencies that can be applied before more expensive multi-agent critique.

5.2 Accountability and Authorship

Concern. The discursive network framework identifies a failure mode it calls *scientific epithesis*, defined as claiming authorship or intellectual credit after only superficial engagement with agent-produced content [22]. This concern connects to longstanding debates about honorary and gift authorship [45]. Current publication ethics guidance is clear that AI tools cannot be authors because they cannot take accountability for research integrity [18–20, 46].

Examples. A researcher uses an agent to draft an entire methods section, makes only cosmetic edits, but claims full authorship. A team lists an agent as a “contributor” without clarifying the nature of its role.

Mitigations.

- Apply ICMJE authorship criteria strictly to human contributors; list AI tools only in methods or acknowledgments sections as appropriate.
- Maintain contribution logs documenting what humans did versus what agents produced.
- Include human accountability statements for all agent-assisted work: “[Author names] take full responsibility for the accuracy and integrity of this work.”
- Ensure that every claim in the final output has been reviewed and affirmed by a responsible human.

Table 1: Common scenarios in AI-assisted scholarly work and their required actions.

Scenario	Acceptability	Required action
Grammar or style polishing on author-drafted text	Usually acceptable	Disclose if the journal or policy requires; otherwise no further action.
AI drafts a paragraph; human verifies, rewrites, and integrates	Acceptable with disclosure	Disclose tool, task, and verification procedure; retain contribution log.
AI writes substantive methods or results; human lightly edits	High risk	Major human re-analysis and rewrite; verify every claim against source; itemize contributions; consider whether the methods are truly the authors’ own work.
AI fabricates citations or numbers	Unacceptable	Reject output; audit the full bibliography or analysis; record the failure in the invalidation log.
AI listed as author or co-author	Unacceptable under COPE/ICMJE/WAME [18–20, 46]	Remove AI from authorship; disclose use in methods or acknowledgments.
Human claims authorship without intellectual contribution	Unacceptable; constitutes scientific epithesis	Reassign credit; document contributions; do not submit.

5.3 Provenance, Auditability, and Reproducibility

Concern. Agentic workflows can involve iterative revisions across multiple models, tools, and sessions. Without systematic logging, it becomes impossible to reproduce results or investigate the source of errors. Reproducibility is a cornerstone of scientific integrity [47, 48], and opaque agent workflows undermine it.

Examples. A researcher cannot reproduce an analysis because they did not record which prompts, model versions, or random seeds were used. An error in a manuscript cannot be traced because intermediate agent outputs were discarded.

Mitigations.

- Maintain tamper-evident records: at minimum, log prompts, outputs, tools called, model versions, and timestamps.
- Store exact prompts and configurations alongside code, seeds, and environment metadata.
- Define handoff points where a human must review and approve outputs before external release.
- Treat agent workflows with the same versioning discipline applied to code: tag releases, document changes, preserve history.
- Record exact model identifiers including snapshot or version strings (e.g., `gpt-4-0613`, `claude-3-5-sonnet-20241022`), not only family names. Where reproducibility is essential and the vendor publishes deprecation timelines, preserve the agent output alongside the prompt so that the workflow can be reconstructed after the underlying model is retired.

5.4 Confidentiality and Privacy

Concern. Sensitive content can leak from agent workflows through several distinct channels: direct transmission to a third-party service at inference time, retention in prompts, logs, memory stores, vector embeddings, or tool-call payloads, and, separately, through model memorization that may surface in later outputs. Membership inference and training-data extraction studies [39, 40] establish that trained models can leak information from their training corpora under some conditions; this is a separate mechanism from the immediate operational risk of pasting protected health information (PHI) or unpublished material into a prompt. Biomedical workflows must address all three channels. Federal agencies have issued explicit guidance prohibiting the use of generative AI in peer review contexts [21, 49].

Examples. A researcher pastes patient identifiers into an agent prompt for help with data cleaning. A grant reviewer uploads proposal text to a commercial LLM to help draft critiques. An agent’s memory retains sensitive information from a previous session.

Mitigations.

- Never submit PHI, unpublished proposals, or confidential peer-review text to any system not approved by the institution for that data class. Where HIPAA applies, require a Business Associate Agreement (BAA) with the vendor.
- Classify data before prompting: public, internal, unpublished, controlled-access, PII, PHI, genomic, tribal or Indigenous, proprietary, peer-review confidential. Match the data class to an approved system before any agent interaction.
- Practice data minimization: redact identifiers using either HIPAA Safe Harbor or Expert Determination [50]; use synthetic or surrogate examples; summarize rather than paste raw records.
- Treat memory stores, tool outputs, logs, and embeddings as governed data repositories subject to the same retention and access-control rules as the underlying data.
- Clear agent state between sessions of different sensitivity levels and verify that vendor retention policies match the data class.
- Verify compliance with applicable IRB protocols, data use agreements (DUAs), the NIH Data Management and Sharing Policy (NOT-OD-21-013) [51], and federal peer-review confidentiality rules before initiating an agent workflow on regulated data.

Policy-sensitive boundary. Using AI to prepare one’s own grant or manuscript is distinct from using AI to review confidential materials. NIH (NOT-OD-23-149) prohibits NIH scientific peer reviewers from using natural-language processors, large language models, or other generative AI to analyze or formulate peer-review critiques for grant applications and R&D contract proposals. NSF prohibits uploading any content from proposals, review information, or related records to non-approved generative AI tools. Both

policies turn on confidentiality; both predate ongoing agency-level revisions; always consult the current agency notice, the signed reviewer agreement, and institutional policy before any AI use in peer review.

5.5 Bias and Fairness

Concern. LLMs can encode and amplify biases present in training data, affecting how populations are characterized, which research questions are prioritized, and how findings are interpreted [10, 52–54]. In biomedical contexts, biased outputs can contribute to health disparities. Multi-agent consensus can suppress valid minority perspectives.

Examples. An agent systematically underrepresents certain demographic groups in literature summaries. A multi-agent workflow converges on stereotyped characterizations of patient populations. An analysis agent defaults to assumptions that are appropriate for majority populations but inappropriate for the population under study.

Mitigations.

- Include diversity-of-critique steps: task at least one agent (or human) with arguing plausible alternatives to the emerging consensus.
- Evaluate outputs for group harm and biased framing when dealing with populations and health disparities.
- Apply established algorithmic fairness frameworks where quantitative assessments are possible.
- Document assumptions about target populations and verify that agent outputs are appropriate for those populations.

Biomedical illustration. Consider an agent that proposes a prediction model using prior healthcare utilization as a proxy for disease burden. The proxy encodes access-to-care inequities: groups with structurally lower access generate fewer utilization events at any given level of disease burden. A model trained on this proxy will systematically underestimate disease in those groups, an artifact of measurement, not biology. The mitigation requires testing subgroup calibration and performance, documenting the assumption that motivated the proxy, and either substituting a less biased target or correcting for differential measurement. Obermeyer et al. [53] document this pattern in a deployed clinical algorithm.

5.6 Security and Adversarial Manipulation

Concern. Agents that browse, retrieve, or execute tools are exposed to adversarial inputs. Prompt injection attacks can cause agents to deviate from intended behavior [55, 56]. Tool-using agents create new attack surfaces (e.g., malicious instructions embedded in retrieved content) [57].

Examples. An agent retrieves a web page containing hidden instructions that override the researcher’s prompt. An agent executes code from an untrusted source, compromising the research environment. An adversary crafts inputs designed to cause the agent to leak sensitive information.

Mitigations.

1. Separate read, reason, and act privileges. Retrieval agents should not execute code or send messages.
2. Apply least-privilege tool scopes. Default to read-only file access; no network egress unless explicitly approved; no production credentials in agent context.
3. Require human approval gates before agents execute code, write files, query databases, send communications, or call external APIs on non-trivial scopes.
4. Sandbox execution in disposable containers with limited filesystem and no uncontrolled network egress. Treat any code retrieved or generated by the agent as untrusted until reviewed.
5. Screen retrieved content. Treat web pages, PDFs, repository files, and emails as untrusted inputs that may contain instructions targeting the agent.
6. Log prompts, outputs, tool calls, model identifiers, environment metadata, and human approvals. Pin package versions and prefer institutional registries when handling sensitive data.

5.7 Sustainability and Equity

Concern. Multi-agent critique workflows impose a compute, latency, and energy burden that grows with ensemble size. In the all-pairs configuration, a workflow with n agents requires approximately n generation passes and $n(n - 1)$ critique passes plus a final harmonization pass; the dominant term is quadratic in n . Architectures that prune to star or hierarchical configurations reduce critique passes to linear in n at the cost of redundancy in detection. Training-cost studies [41, 42] motivate the broader sustainability concern, but inference-specific estimates should be used when making quantitative claims. When best practices require extensive verification pipelines, resource constraints can create inequities: well-funded labs can afford thorough verification while resource-limited researchers cannot.

Examples. A research protocol calls for seven-agent cross-critique on all outputs, making the workflow prohibitively expensive for many groups. A lab in a low-resource setting must choose between thorough verification and operational feasibility.

Mitigations.

- Right-size verification: scale agent counts and critique depth to the stakes and complexity of the task.
- Prefer cheap checks first: use validators, linters, unit tests, and retrieval before adding inference-heavy critique agents.
- Track and report compute usage; justify multi-agent overhead for high-risk outputs.
- Consider verification cost as a factor in workflow design, not an afterthought.

6 Governance-by-Design Principles

The preceding risk analysis suggests that responsible agent use requires proactive design, not merely reactive correction. We propose a set of governance-by-design principles, along with practical tools for implementation.

6.1 Core Principles

1. **Human accountability is non-delegable.** Regardless of agent contributions, human researchers bear full responsibility for the accuracy, integrity, and ethical compliance of their work. Agents are tools, not collaborators in the accountability sense.
2. **Transparency scales with stakes.** The depth of documentation, verification, and disclosure should be proportional to the potential impact of errors. Routine internal work requires less formality than peer-reviewed publications or clinical applications.
3. **Verification is part of the workflow.** Agent outputs should not be treated as finished products. Every workflow should include explicit verification stages appropriate to the output type.
4. **Privacy protection is a precondition.** Sensitive data should never enter agent workflows without appropriate safeguards. When in doubt, exclude.
5. **Reproducibility requires records.** If a workflow cannot be reconstructed from available logs, it has not met the reproducibility standard expected of scientific research.

6.2 Practical Checklist

Before deploying agents in a biomedical workflow, researchers should confirm:

1. **Scope:** What decisions and actions are agents permitted to take? What requires human approval?
2. **Data:** Does the workflow involve PHI, unpublished proposals, or confidential peer-review material? If so, is the agent system approved for such data?
3. **Provenance:** Are prompts, outputs, tool calls, and model versions logged in a retrievable format?
4. **Verification:** Which output types require independent verification? Who will verify, and by what method?
5. **Authorship:** Who meets authorship criteria for outputs of this workflow? How will AI assistance be disclosed?
6. **Bias:** Have outputs been evaluated for biased framing and potential group harms?
7. **Security:** Are tools sandboxed? Are external resources filtered? Are permissions minimized?
8. **Sustainability:** Is the verification overhead justified for the risk level? Has compute cost been considered?

An expanded version of this checklist, with detailed questions for each category, is provided in Appendix [A](#).

6.3 Disclosure Template

For manuscripts and other formal outputs, we recommend a disclosure statement of the following form:

We used AI-assisted tools [specify names and versions] for [specify tasks: e.g., literature search, code generation, language editing, summarization]. All outputs were reviewed and verified by the authors, who take full responsibility for the accuracy, integrity, and originality of the work and for compliance with all applicable data governance and ethical requirements.

This template can be adapted to specific journal requirements and expanded to include additional detail as appropriate. Appendix [B](#) provides three sample disclosure statements calibrated to different levels of agent involvement.

7 An Illustrative Workflow

To make the framework concrete, we trace one analytical task end-to-end. The task is a literature triage for a county-level study of underweight birth rates in Texas using historical National Center for Health Statistics (NCHS) natality data. The workflow uses two agents: a retrieval agent that searches the published literature, and an analysis agent that drafts a summary memo. Both agents are tool-using LLMs operating under the governance protocol described in Section [6](#).

Factual invalidation, detected and corrected. The retrieval agent returned a paragraph citing a 2019 paper that does not exist. Verification step: every DOI in the agent output was checked against the publisher’s resolver before any text was retained. The fabricated citation was removed; the bibliography was audited for additional fabrications and one further was found.

Logical invalidation, detected and corrected. The analysis agent drafted a sentence asserting that the underweight birth rate was significantly higher in a county after a stated odds ratio. The odds ratio was for the maternal age covariate, not the county effect. Verification step: every statistical claim in the draft was traced to its source table. The sentence was rewritten.

Normative invalidation, detected and corrected. The analysis agent drafted a sentence describing racial disparities in birth outcomes as “intrinsic to the population.” Verification step: a normative review by a human author replaced the framing with social-determinants language consistent with the published literature on racism as a structural cause of disparate outcomes.

Structural invalidation, detected and corrected. The analysis agent produced a CSV summary in which the column order did not match the schema of the downstream visualization tool. Verification step: schema validation on the CSV failed at the boundary, and the agent was prompted to regenerate the file in the correct order.

Final disclosure. The disclosure statement for the resulting memo follows the template in Appendix B and itemizes the verification procedures applied at each stage.

This single workflow illustrates all four invalidation types and the corresponding verification strategies. In practice, not every workflow surfaces every type; documenting which types did not appear is itself a useful component of the invalidation log.

8 Discussion

8.1 Relationship to Existing Frameworks

The framework presented here complements rather than replaces existing guidance on AI ethics and research integrity. The NIST AI Risk Management Framework [14, 58] provides general principles for AI governance; our contribution is to operationalize those principles for the specific context of biomedical research workflows. Similarly, the UNESCO Recommendation on AI Ethics [16] and OECD AI Principles [17] articulate high-level values; we translate those values into concrete practices.

Our focus on agents addresses a gap in current publication ethics guidance. The positions from COPE, ICMJE, and WAME [18–20, 46] primarily address disclosure of AI assistance in manuscript preparation. We extend these considerations to the full research workflow, including data analysis, code development, and internal documentation.

8.2 Limitations

Several limitations should be acknowledged:

1. **Rapid technological change.** AI agent capabilities are evolving quickly, and some of the specific risks and mitigations discussed here may become obsolete as systems improve or new vulnerabilities emerge.
2. **Context dependence.** The appropriate level of verification and documentation depends on factors (institutional resources, regulatory environment, research domain) that vary across settings. Our framework provides principles rather than rigid prescriptions.
3. **Empirical validation.** While our framework draws on established research on LLM reliability and agent architectures, the specific efficacy of proposed mitigations awaits empirical evaluation in diverse biomedical contexts.
4. **Scope of “agent.”** The boundary between agentic and non-agentic AI use is fuzzy. A researcher who iteratively prompts a chatbot and manually integrates results is engaging in a form of human-agent collaboration not cleanly captured by our definitions.

8.3 Future Directions

Several directions merit further investigation:

- Development of standardized logging formats and tools for agent workflow provenance.
- Empirical studies of invalidation rates across biomedical tasks and agent architectures.
- Investigation of techniques for detecting and mitigating bias in multi-agent consensus.
- Integration of agent ethics training into biomedical research curricula and RCR programs.
- Regulatory frameworks specific to agent-assisted clinical research and healthcare applications [59, 60].

9 Conclusion

LLM-based AI agents offer substantial potential benefits for biomedical data science, from accelerating routine tasks to enabling new analytical approaches. Realizing these benefits responsibly requires extending our ethical frameworks to address the distinctive features of agentic workflows: autonomous action, persistent state, multi-step reasoning, and multi-agent coordination.

This article has presented such an extension, grounded in the concepts of invalidation (a broader framing than hallucination) and multi-agent critique (with its attendant tradeoffs). We have mapped seven categories of ethical risk and provided concrete mitigations for each. Our governance-by-design principles and practical tools offer a path toward responsible adoption.

The core message is not that agents should be avoided, but that they should be used with deliberate attention to the expanded ethical surface they create. Human researchers remain accountable; agents are tools that require oversight, verification, and documentation commensurate with their autonomy and the stakes of the research. By building ethical considerations into workflow design from the outset, the biomedical research community can harness the capabilities of AI agents while upholding the integrity standards that scientific progress requires.

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Competing Interests

The author declares no competing interests.

Data Availability

This article does not report primary data. All referenced sources are cited in the bibliography.

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A Governance Checklist (Expanded)

Category	Questions to Address
Scope Definition	What tasks will agents perform? What decisions require human approval? What are the boundaries of agent autonomy? Are there any categories of action that agents should never take?
Data Classification	Does the workflow involve PHI, PII, or other protected data? Are there unpublished findings or confidential peer review materials? Is the agent system approved for the data sensitivity level? Have all applicable data use agreements been reviewed?
Provenance Logging	Are prompts, outputs, and tool calls logged? Are model versions and configuration parameters recorded? Are timestamps captured? Can the workflow be reconstructed from available logs?
Verification Protocol	Which output types require independent verification? Who is responsible for verification? What methods will be used (retrieval, recomputation, expert review)? What is the verification budget (time, resources)?
Authorship and Attribution	Which humans meet authorship criteria? How will AI assistance be disclosed? Is there a contribution log distinguishing human and agent work? Who takes accountability for each section or claim?
Bias Assessment	Have outputs been evaluated for biased framing? Are there potential harms to specific populations? Has diversity of perspective been incorporated into the workflow? Are assumptions appropriate for the target population?
Security Controls	Are tools sandboxed with minimal permissions? Are external resources filtered through allowlists? Is retrieved content screened before agent processing? Are there monitoring and kill-switch mechanisms?
Sustainability Considerations	Is the verification overhead proportionate to risk? Has compute cost been estimated and justified? Are there opportunities to reduce agent count or critique depth without compromising quality?

B Sample Disclosure Statements

B.1 Minimal Disclosure (Low Agent Involvement)

We used [tool name] for [specific limited task, e.g., grammar checking, code formatting]. All substantive content was authored by the listed authors.

B.2 Standard Disclosure (Moderate Agent Involvement)

We used GPT-4 (gpt-4-0613, OpenAI) and Claude 3.5 Sonnet (claude-3-5-sonnet-20241022, Anthropic) for literature search and initial code drafting. No protected health information, controlled-access data, confidential peer-review material, or unpublished third-party material was provided to either tool. All outputs were reviewed, verified, and revised by the authors, who take full responsibility for the accuracy, integrity, originality, source attribution, and data-governance compliance of the work. Verification procedures included source checking against retrieved DOIs, recomputation of all quantitative claims, code review by a second author, and a bias audit on subgroup outputs.

B.3 Detailed Disclosure (Substantial Agent Involvement)

This work made substantial use of AI-assisted tools for [specific tasks]. Specifically, [Tool A, version X] was used for [task 1]; [Tool B, version Y] was used for [task 2]. The authors implemented the following verification procedures: [describe]. Complete logs of agent interactions are available [location/upon request]. All claims, analyses, and conclusions were reviewed and affirmed by the authors, who take full responsibility for the work. Data classes provided to each tool and verification procedures applied to each output category must be documented.

C Assessment Instrument

The following assessment can be used in educational settings to evaluate understanding of agent ethics concepts.

C.1 Knowledge Check (8 points)

1. (1 pt) Which feature most distinguishes an AI agent from a single-turn chatbot?
 - A. Larger parameter count
 - B. Ability to plan and call tools
 - C. Better grammar
 - D. Faster response time
2. (1 pt) Which of the following are examples of invalidation? Mark every option that applies.
 - A. A fabricated citation.
 - B. A statistically invalid inference from correct intermediate results.
 - C. Malformed JSON passed to a downstream pipeline.
 - D. Disclosure of protected health information in an output.

Answer key: A, B, C, and D each independently. Full credit requires marking all four.

3. (1 pt) According to COPE/ICMJE/WAME guidance, AI tools should be listed as authors:
- A. Always
 - B. Only if they wrote more than 50% of text
 - C. Never
 - D. Only in methods papers
4. (1 pt) True or False: A principal investigator may claim authorship on a methods section drafted entirely by an LLM if the PI has not reviewed each claim. *Answer:* False. Credit answers that cite the ICMJE accountability requirement.
5. (2 pts) You are serving on a federal grant review panel. Which actions are policy problematic? Mark all that apply.
- A. Taking handwritten notes on the proposals.
 - B. Uploading proposal text into a public LLM to draft critiques.
 - C. Discussing the proposal in detail with a colleague at the same institution who is not on the panel.
 - D. Checking formatting rules in the solicitation.

Answer: B and C. B violates NIH NOT-OD-23-149 and NSF confidentiality. C violates standard reviewer confidentiality agreements.

6. (2 pts) A multi-agent “all-pairs critique” workflow with $n = 7$ agents most directly raises which concerns? Mark all that apply.
- A. Compute and energy cost growing approximately as n^2 .
 - B. Loss of architectural diversity if all agents come from the same model family, leading to correlated errors.
 - C. Privacy amplification when sensitive content is shared with multiple services.
 - D. Guaranteed absence of bias from the consensus output.

Answer: A, B, and C. D is false; consensus among similar models does not guarantee correctness.

C.2 Short Response (2 points)

7. (1 pt) Write a one-sentence disclosure describing how AI tools were used in a manuscript or analysis you might produce.
8. (2 pts) Name two safeguards you would implement before allowing an agent to execute code on biomedical data, and justify each in one sentence. *Rubric:* Full credit for two of the following with one-sentence justification: sandboxed disposable container, human approval gate before write or network operations, audit logging of all tool calls, data redaction or synthetic substitution, restricted network egress, no production credentials in agent context.

C.3 Answer Key (Multiple-Choice Summary)

- Q1: B. Q2: A, B, C, D (all four). Q3: C. Q4: False. Q5: B and C. Q6: A, B, and C. Q7: Credit for specific mention of tool and task. Q8: See rubric in question.